

Gabrielle Johnson's Deployment System Requirements

(a) Deployment vision

The nurse carries a handheld device that shows a suggested ESI level and the reasons behind it. Patients wear a watch that takes their vitals and keeps checking them during the wait. A kiosk handles check-in and lets patients see where they are in the queue.

The plan is to build a prototype, test it, use what we learn to retrain the model on Mercer's own patients, test it again, and only then roll it out.

Stage	What happens	Closing
Alpha test (8–12 weeks)	The system captures vitals, complaints, the nurse's final level — but the model's suggestion is hidden. Nurses use it as a data-capture tool only	At least 1,000 Mercer patients captured
Recalibration (2–4 weeks)	The model is retrained on the Mercer data collected in alpha, and re-measured against Mercer's own nurse decisions	The model performs at least as well on Mercer patients as it did on the original dataset
Beta test (6–8 weeks)	Suggestions are shown and deterioration alerts are switched on, running alongside normal triage. The nurse records her own level first, then sees the suggestion, so we can measure how often they agree	Agreement is stable, false alarms stay within the limit agreed, and nurses say it is quicker and more accurate than paper
Full rollout	The suggestion appears before the nurse commits; the system becomes the normal way of working	Reviewed every three months for accuracy and for fairness across patient groups

(b) System requirements

Functional — what the system must do

ID	Requirement
F1	For patients aged < 18 or obstetric, MUST NOT generate a level
F2	MUST send a deterioration alert to the nurse handheld device with the patient's location/name and a named owner, escalating to the charge nurse if unacknowledged within 120 s
F3	A checked-in but unreviewed patient MUST escalate on expiry of the Mercer ESI target (≤ 10 min ESI 2, ≤ 30 min ESI 3, ≤ 60 min ESI 4)
F4	The watch MUST send a heartbeat every 60 s ; three consecutive misses raise a device-failure alert
F5	An interrupted assessment MUST resume with zero re-entry of values already entered
F6	No patient-facing surface MUST EVER show an ESI level, a numeric vital, a clinical reason or a countdown
F7	MUST store every suggestion, final ESI level and override reason with a timestamp and owner, and generate a report rate weekly

Non-functional — the conditions it must do it under

ID	Requirement
NF1	The suggestion and its reasoning MUST appear within < 60 s of vitals confirmed
NF2	A deterioration alert MUST reach the handheld device within 10 s of the triggering sample
NF3	MUST degrade to manual entry within 15 s of a device-stream failure

NF4	On a 360 × 740 px display, at least one contradicting feature and the blind-spot card MUST be visible without scrolling, with ACCEPT positioned below them
NF5	MUST remain legible under fluorescent clinical lighting at up to 1 m, with text contrast $\geq 4.5:1$, and every state distinguishable without colour by icon, label and position
NF6	Deterioration alerts MUST be distinguishable from routine traffic by vibration alone
NF7	The strap MUST be removable by staff
NF8	Capture and queue state MUST survive ≥ 30 min without a network and catch up with nothing lost
NF9	Location data MUST be deleted at the end of the encounter; race, ethnicity, insurance, religion, marital and employment status and language MUST NOT be model inputs but MUST stay available for fairness checks

Integration — what it must connect to

ID	Requirement
I1	MUST run standalone with no hospital IT dependency, holding its own structured store
I2	MUST receive vitals from the patient watch over BLE, presented for nurse accept/reject before reaching the model
I3	MUST resolve indoor location from BLE/UWB anchors to seating zone level across the waiting room, fast-track area and corridors
I4	MUST deliver escalations to the charge nurse's handheld device as a separate destination from the assigned nurse's

I5	MUST fall back to conventional paper triage with no data loss when unavailable
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(c) Inputs and outputs

What goes in. Vitals (HR, SpO₂, respiratory rate, temperature, etc) stream from the patient's watch, and the nurse accepts or rejects each before it reaches the model. Chief complaint comes from a kiosk picklist or the nurse; free text is not accepted. Age, sex and arrival mode are typed in, and age decides whether the model runs at all.

Outputs the model emits.

Output	Destination	Form
Suggested ESI level, reasoning,	Nurse handheld device only	A number 1–5 with its target time-to-be-seen, plus ≤ 3 supporting and ≤ 2 contradicting features as labelled bars, and amber chips on low-confidence values
Blind-spot warning / suppression notice	Nurse handheld device only	Amber card for abnormal/low confidence values;
Ambient attention state	Patient watch	Steady amber + icon + word + haptic. No level, number or reason
Clinical alert	Nurse handheld device	Unique vibration; trend, location, named owner, escalation countdown
Escalation	Charge nurse handheld device	On unacknowledged alert (120 s) or expired ESI target
Status	Kiosk, on watch scan	Queue position or range

(d) Build path and justification

Chosen: an installed mobile app

1. **It has to be mobile, because Mercer has no staffed triage desk.** Nurses are constantly on the go, so a pager style device, which is close to something they already carry and understand, will be easier to fit into how they actually work.

2. **Installed rather than web, because four requirements need it.** It must be an installed app rather than a web app because: web Bluetooth and the Vibration API are not supported in iOS Safari; browser storage can be removed under pressure; and a web app cannot reliably wake in the background to deliver a deterioration alert.
3. **The physical devices come last.** The physical devices should come last. A wearable device that informs triage decisions will probably be regulated as a medical device, while purchasing validated sensors and installing location beacons could take months. Starting there would commit the most money to the slowest and most heavily regulated part of the system before establishing whether nurses will use the interface.